Venkata Lakshmi Parimala Pasupuleti

AI Engineer · Agentic AI · RAG · LLM Evaluation · Guardrails · Async Processing · Python · FastAPI · AWS

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SUMMARY

AI engineer specializing in agentic systems, RAG pipelines, LLM evaluation, and production backend services. Currently at Intuit building Anthropic Claude integrations into production financial services - tool use, retrieval grounding, structured outputs, hallucination mitigation, guardrails, and LLM-as-judge evaluation harnesses with CI regression gating. Built **Leasel** - an 8-agent LangGraph customer agent platform with tool calling, durable state, context management, multi-step workflows, and HITL escalation (93.6% precision, 1,240 test cases). Built **AgentProbe** - an open-source LLM evaluation platform with offline golden-dataset evals, LLM-as-judge scoring, online quality metrics, and CI regression harness gating releases on eval thresholds. At Entrata (7+ years, 100K+ users), built and operated RabbitMQ-based async distributed processing for rent updates across thousands of tenant records - retry logic, idempotency guarantees, and failure routing in a regulated multi-tenant SaaS. Strong across Python, FastAPI, async/event-driven systems (RabbitMQ, SQS), AWS, Docker, and Kubernetes. M.S. Data Analytics Engineering, George Mason University (GPA 3.87).

CORE COMPETENCIES

Agentic AI · Tool Use · Context Management · Orchestration

RAG Pipelines · Retrieval · Grounding · Hallucination Mitigation

Structured Outputs · Prompt Design · Multi-Step Workflows

LLM Evaluation · LLM-as-Judge · Golden Datasets · CI Regression

Error Analysis · Quality Metrics · Production Signals

Guardrails · Fallbacks · Observability · Failure Handling

Async Processing · RabbitMQ · SQS · Kafka · Event-Driven

Distributed Systems · Idempotency · Retry Logic

Python · FastAPI · Django · TypeScript · Node.js

AWS (Lambda · SQS · S3 · RDS · API Gateway) · Docker · Kubernetes

PostgreSQL · Redis · ChromaDB · SQLite

LangGraph · LangChain · Anthropic Claude · OpenAI

GitHub Actions CI/CD · OpenTelemetry · Prometheus · Grafana

EXPERIENCE

Software Engineer

Jun 2026 - Present

Intuit — Virginia, USA

- **Agentic AI in production:** Integrate Anthropic Claude into Python/FastAPI financial services - tool use with retrieval grounding, context management for multi-step workflows, structured-output schemas, and hallucination mitigation via RAG-based citation grounding; guardrails with input validation, sensitive-data handling, and fallbacks for reliable production behavior.
- **LLM evaluation and quality iteration:** Build LLM-as-judge evaluation harnesses with golden datasets and CI regression gates; perform qualitative error analysis on production failures and turn them into evals that catch that class of failure next time; monitor shipped experiences via production signals and quality metrics to guide iteration.
- Build ML data pipelines (scikit-learn, XGBoost); React/TypeScript front ends; containerize services with Docker through GitHub Actions CI/CD to AWS with structured logging and CloudWatch monitoring.

Senior Software Engineer

Oct 2020 - Jul 2024

Entrata — Apartment & Affordable Housing SaaS (regulated multi-tenant, 100K+ monthly active users)

- **Async distributed processing at scale:** Built and owned RabbitMQ-based async message bus processing for rent updates across thousands of tenant records - retry logic, idempotency guarantees, dead-letter routing, and failure handling; owned through production incidents and annual schedule changes.
- Owned HUD income-limit compliance ETL pipeline for 6 years (180K records, zero compliance violations); PHI-compliant AWS S3 migration (Libsodium encryption, 100% coverage); Docker microservices redesign (50% scalability improvement, 30% faster deployments); optimized 200+ SQL queries (30% performance gain).
- Python/Bash automation eliminating 80% of manual workload; internal debugging tooling cutting triage time by 40%; mentored 5+ engineers; 300+ code reviews across 120+ Agile sprints.

Junior Software Engineer

Sep 2018 - Oct 2020

Entrata India — Pune

- Built backend modules (Python, PHP, Java, PostgreSQL); delivered large-scale data integration workflows (1.6M records); led AWS S3 migration from legacy storage.

PROJECTS

Leasel - Customer Agent Platform: Tool Use · RAG · Multi-Step Workflows 2026 | Python, LangGraph (8-agent), FastAPI, ChromaDB, CI/CD | github.com/venkatalakshmiparimala/Lease_complaince_rag

- Production customer agent platform: **8-agent LangGraph workflow** with tool/function calling, durable context management, multi-step orchestration, and conditional routing; **RAG pipeline** (OCR ingestion, hybrid search, TF-IDF + semantic reranking, citation grounding for hallucination mitigation); **guardrails** routing regulated decisions to humans; **CI eval harness** with LLM-as-judge and golden datasets: 93.6% precision, 87.9% F1 across 1,240 test cases.

AgentProbe - LLM Evaluation Infrastructure & Quality Metrics 2025 | Python, FastAPI, PostgreSQL, CI/CD | github.com/venkatalakshmiparimala/AgentProbe

- Open-source LLM eval platform: **offline evals** with golden datasets and LLM-as-judge; **online quality metrics** tracking accuracy over time; **CI regression harness** gating releases on eval thresholds; 5-metric eval engine (accuracy, hallucination rate, retrieval quality, completeness, compliance); full tracing and structured logging per agent run. Used as a production release gate before every AI deploy.

AI on FHIR — Clinical Conversational AI & Tool Use 2024 | Python, Flask, React, BERT, LLMs, FHIR REST APIs, Docker | github.com/venkatalakshmiparimala/AI_on_FHIR

- Translates natural-language questions into FHIR REST API calls using LLMs, BERT, and embeddings - cut clinical lookup time ~60%. HIPAA-style RBAC, OAuth-secured API integration, containerized with Docker.

EDUCATION

M.S. Data Analytics Engineering — George Mason University, Virginia | GPA 3.87 May 2026

B.Tech Computer Science & Engineering — JNTU Anantapur 2017